

MACHINE LEARNING WEEK 10

LAB REPORT

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PES2UG23CS381

Course: UE23CS352A

12-10-2025

PART 1:

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SVM with LINEAR Kernel <PES2UG23CS381>
      precision    recall  f1-score   support

     0       0.85      0.89      0.87        75
     1       0.89      0.84      0.86        75

 accuracy      0.87      0.87      0.87       150
 macro avg      0.87      0.87      0.87       150
 weighted avg      0.87      0.87      0.87       150

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SVM with RBF Kernel <PES2UG23CS381>
      precision    recall  f1-score   support

     0       0.95      1.00      0.97        75
     1       1.00      0.95      0.97        75

 accuracy      0.97      0.97      0.97       150
 macro avg      0.97      0.97      0.97       150
 weighted avg      0.97      0.97      0.97       150

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SVM with POLY Kernel <PES2UG23CS381>
      precision    recall  f1-score   support

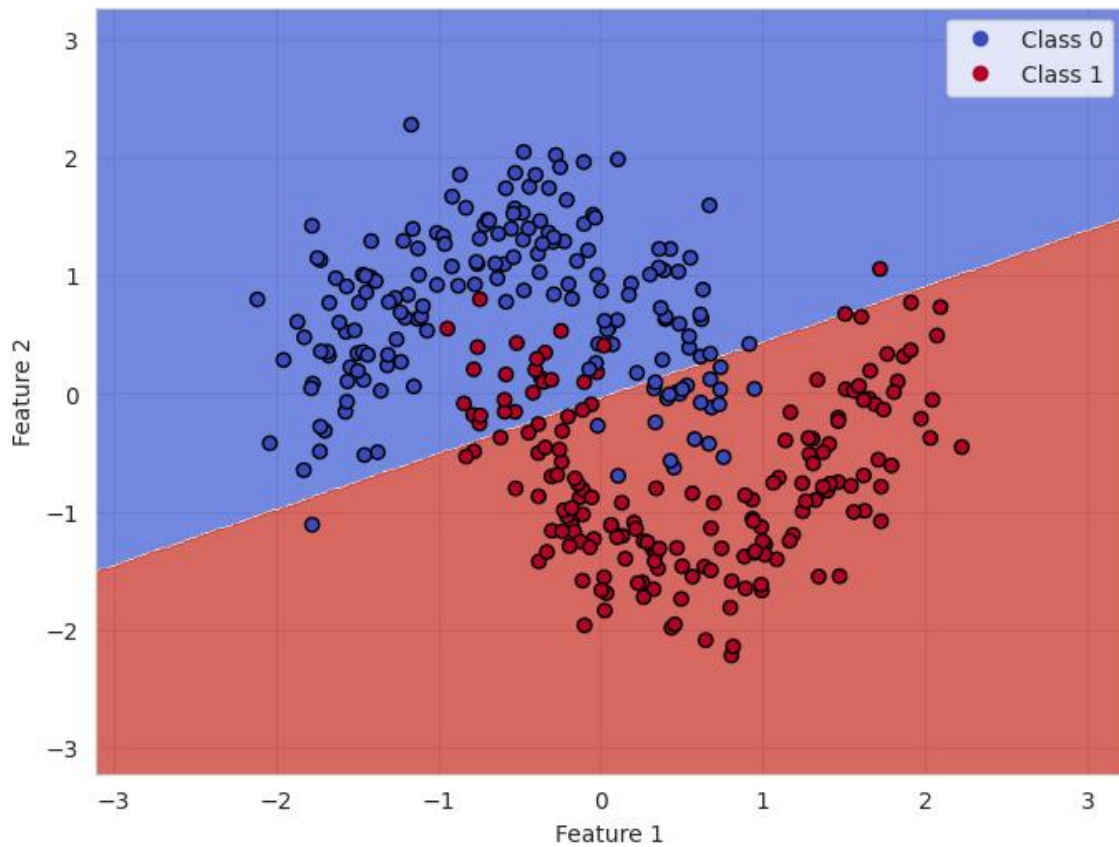
     0       0.85      0.95      0.89        75
     1       0.94      0.83      0.88        75

 accuracy      0.89      0.89      0.89       150
 macro avg      0.89      0.89      0.89       150
 weighted avg      0.89      0.89      0.89       150

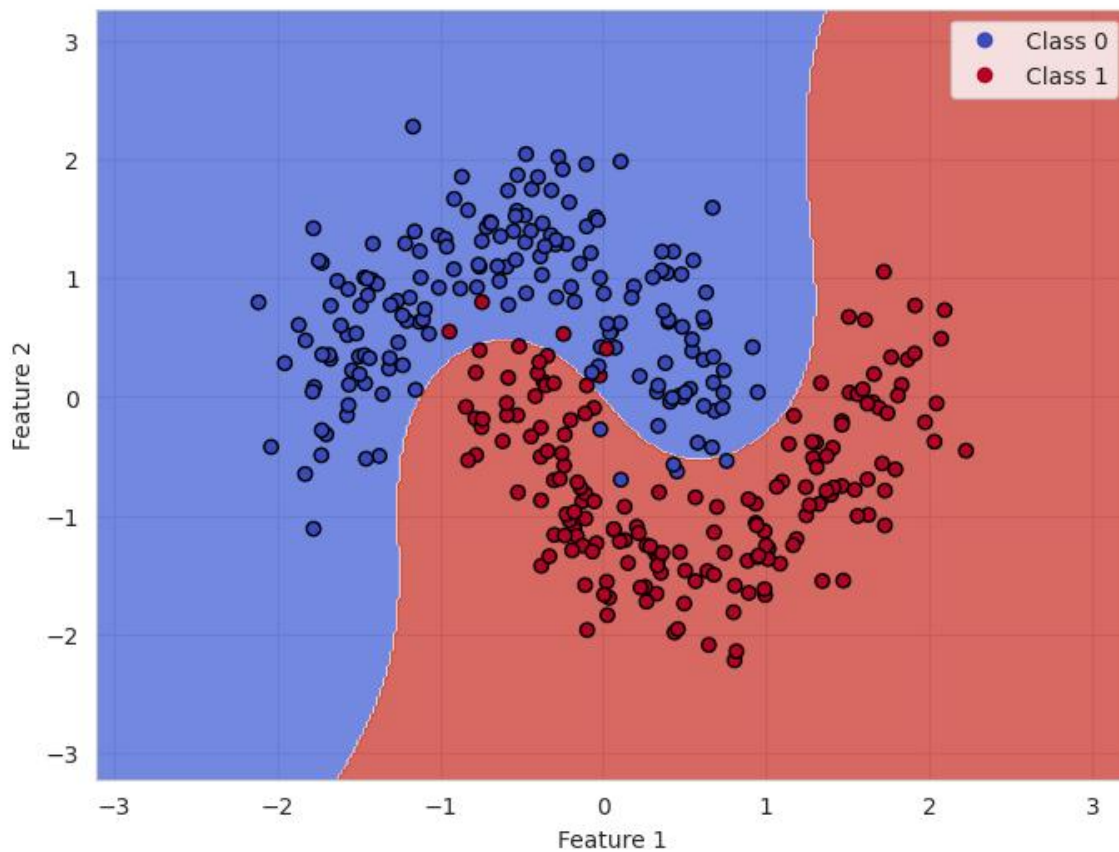
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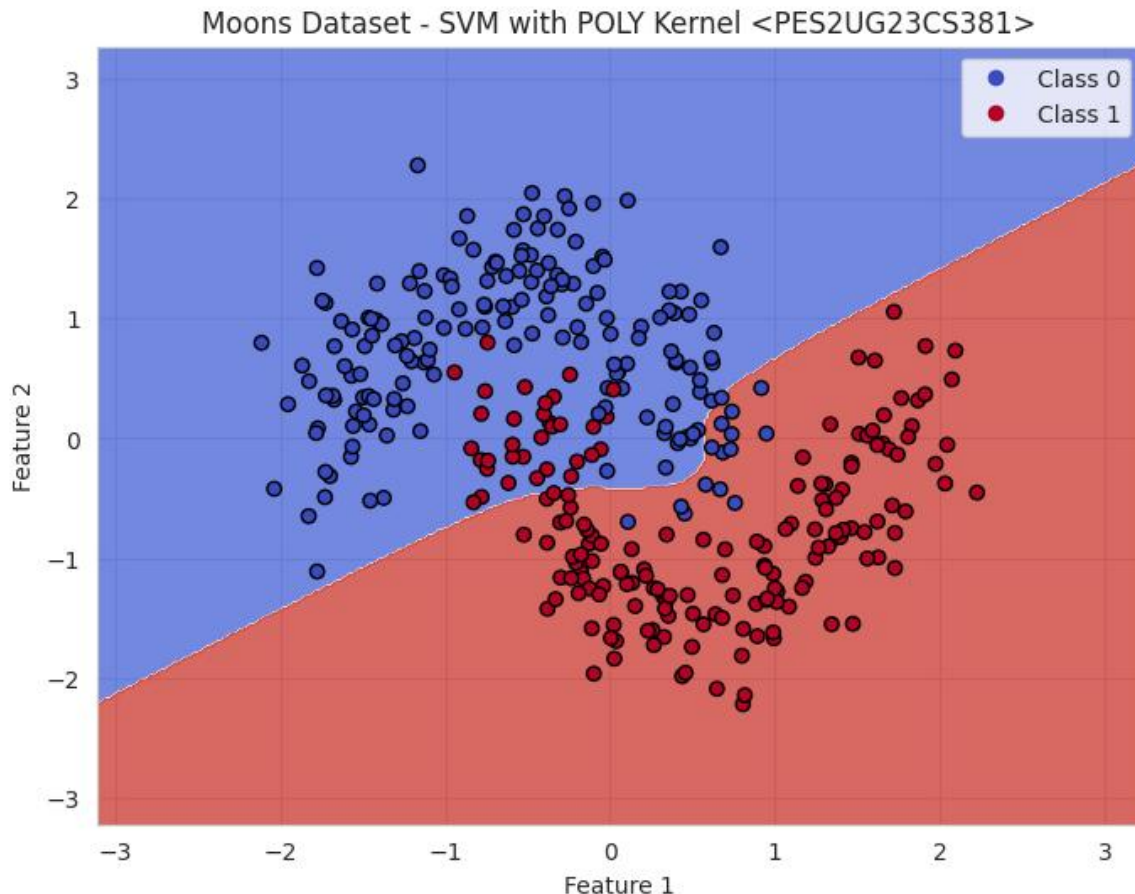
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Moons Dataset - SVM with LINEAR Kernel <PES2UG23CS381>



Moons Dataset - SVM with RBF Kernel <PES2UG23CS381>





Analysis Questions

1. Based on the metrics and the visualizations, what inferences about the performance of the Linear Kernel can you draw?

Based on the visualization and metrics, the Linear Kernel is **poorly suited** for the Moons dataset. The core issue is that the Moons dataset is inherently non-linear, with its data points arranged in two interlocking crescent shapes. A linear kernel, can only create a straight-line decision boundary. The visualization clearly shows this limitation; the straight line is incapable of separating the two "moon" clusters effectively, leading it to cut through both and misclassify a significant number of points. This poor fit is quantified by the metrics. While an accuracy of 87% is not a complete failure, it is the lowest among the three tested kernels. This reflects the visual evidence that the model's fundamental assumption of linear separability does not hold for this data structure, making it a suboptimal choice.

2. Compare the decision boundaries of the RBF and Polynomial kernels. Which one seems to capture the shape of the data more naturally?

The **RBF kernel** captures the shape of the data more naturally and effectively than the Polynomial kernel.

- Polynomial Kernel: This kernel produces a simple, gentle curve. While this is an improvement over the straight line of the linear kernel, the boundary is not flexible enough to trace the distinct crescent shape of the data clusters. It results in a number of visible misclassifications and achieves an accuracy of 89%.
- RBF Kernel: The RBF kernel creates a much more complex and adaptable non-linear boundary. The visualization shows its decision boundary is a tight, smooth curve that almost perfectly follows the natural separation between the two moons. This superior fit is strongly supported by its accuracy of 97%, indicating it makes very few errors.

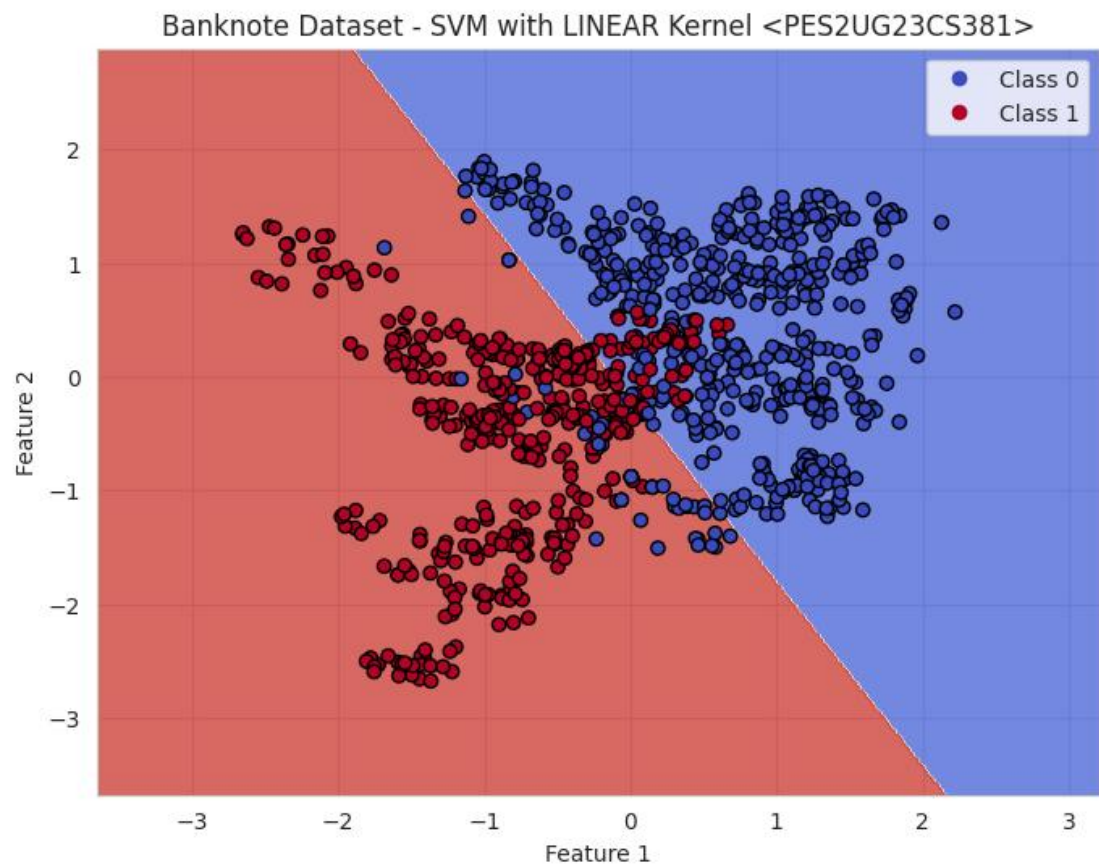
In conclusion, the RBF kernel's flexibility allows it to model the intricate, non-linear structure of the Moons dataset with high precision, making it the most effective model of the three.

PART 2:

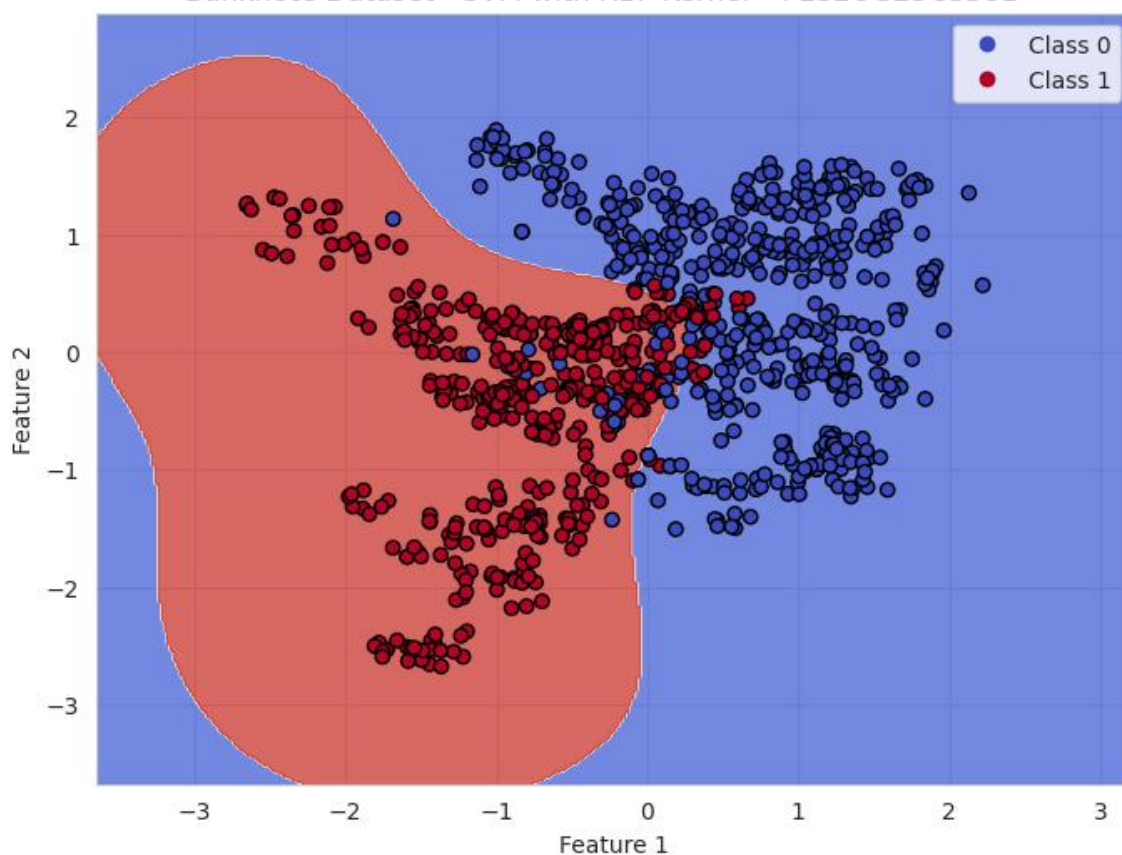
SVM with LINEAR Kernel <PES2UG23CS381>				
	precision	recall	f1-score	support
Forged	0.90	0.88	0.89	229
Genuine	0.86	0.88	0.87	183
accuracy			0.88	412
macro avg	0.88	0.88	0.88	412
weighted avg	0.88	0.88	0.88	412

SVM with RBF Kernel <PES2UG23CS381>				
	precision	recall	f1-score	support
Forged	0.96	0.91	0.94	229
Genuine	0.90	0.96	0.93	183
accuracy			0.93	412
macro avg	0.93	0.93	0.93	412
weighted avg	0.93	0.93	0.93	412

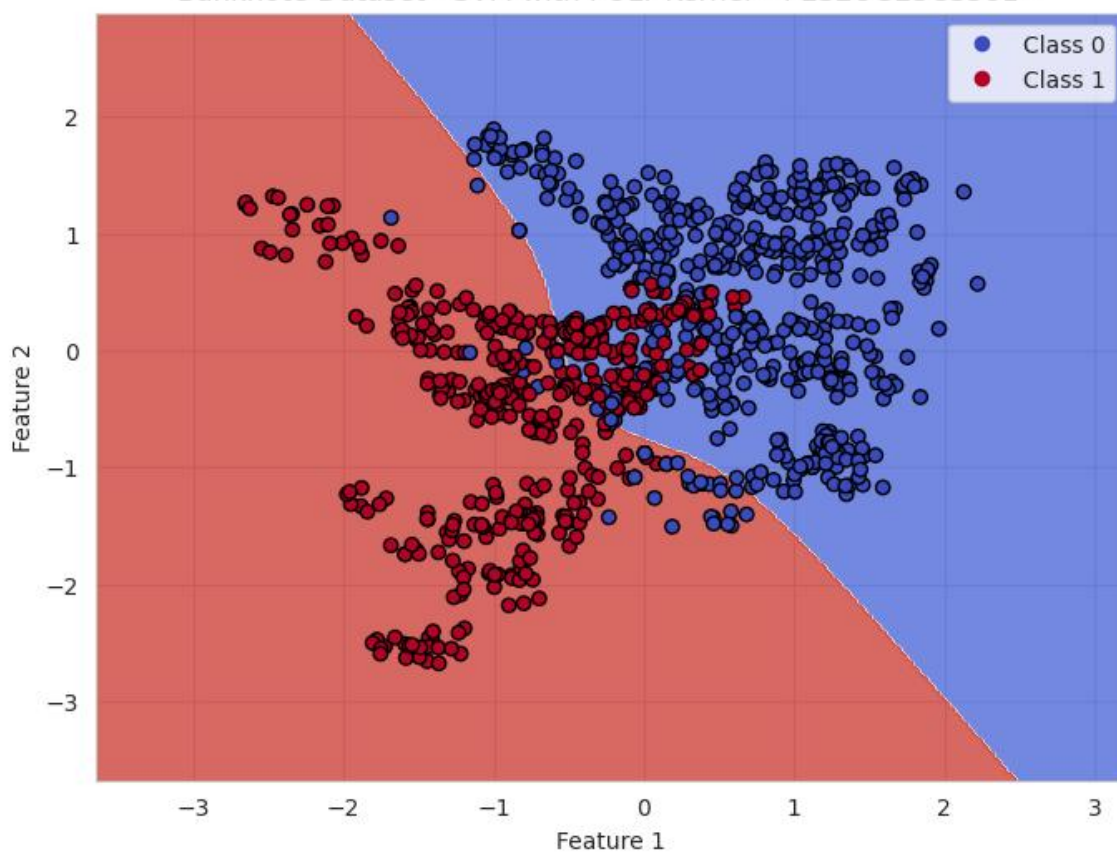
SVM with POLY Kernel <PES2UG23CS381>				
	precision	recall	f1-score	support
Forged	0.82	0.91	0.87	229
Genuine	0.87	0.75	0.81	183
accuracy			0.84	412
macro avg	0.85	0.83	0.84	412
weighted avg	0.85	0.84	0.84	412



Banknote Dataset - SVM with RBF Kernel <PES2UG23CS381>



Banknote Dataset - SVM with POLY Kernel <PES2UG23CS381>



Analysis Questions

1. In this case, which kernel appears to be the most effective?

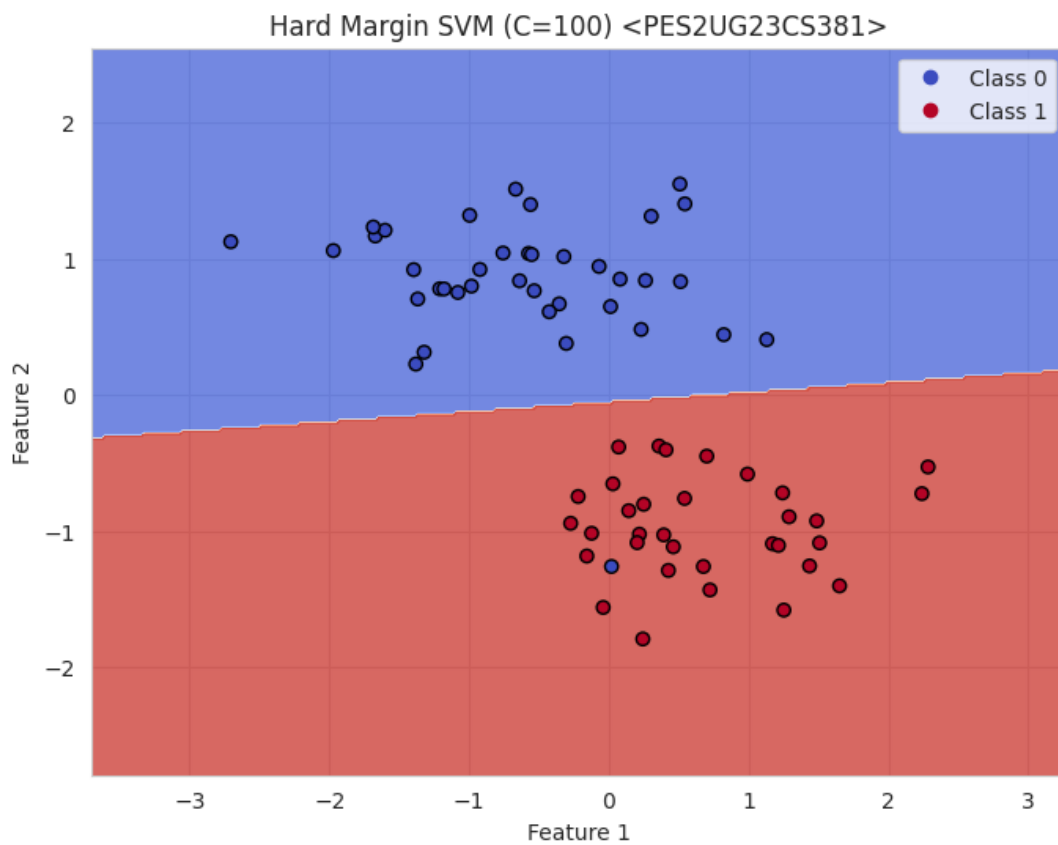
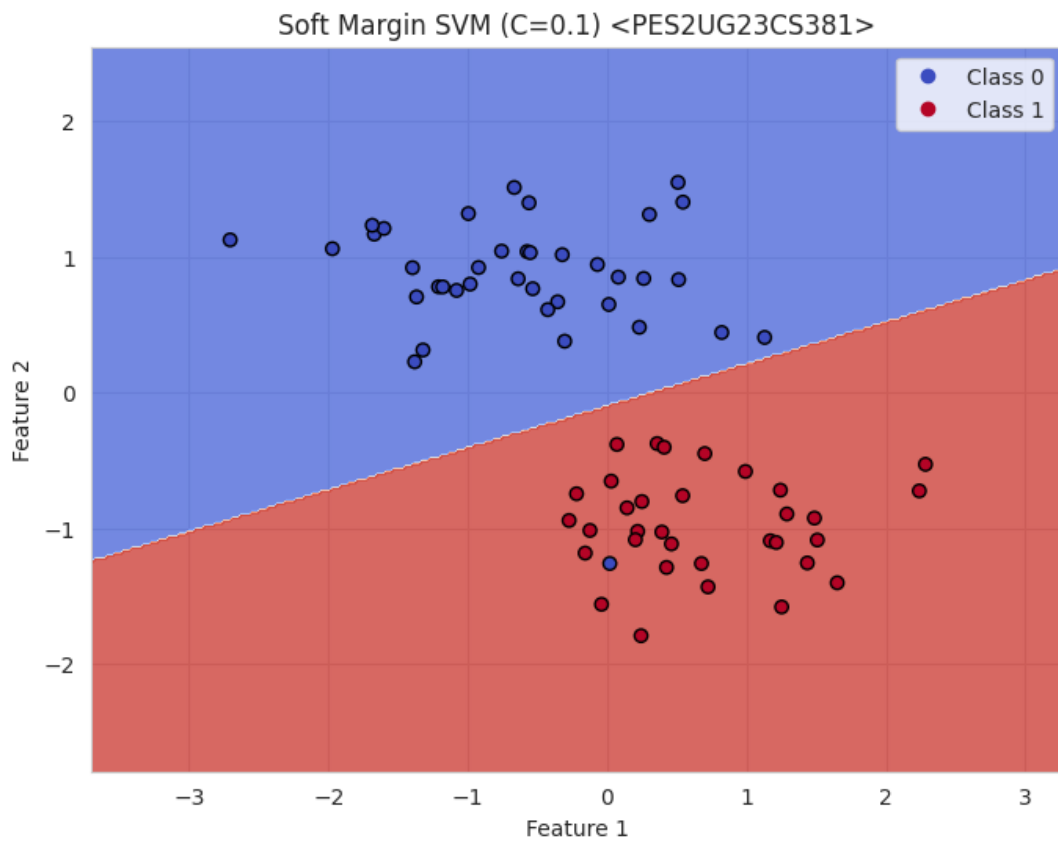
The **RBF kernel** appears to be the most effective for the Banknote Authentication dataset. It achieves the highest overall accuracy of 93%, outperforming both the Linear (88%) and Polynomial (84%) kernels. Furthermore, it scores the highest on the macro average for precision, recall, and f1-score (0.93), indicating a balanced and superior performance in correctly classifying both "Forged" and "Genuine" banknotes. The visualization supports this, showing a flexible decision boundary that effectively isolates the two classes.

2. The Polynomial kernel shows lower performance here compared to the Moons dataset. What might be the reason for this?

The Polynomial kernel's lower performance on the Banknote dataset compared to the Moons dataset likely stems from the **different geometric structures** of the data.

The Moons dataset is intentionally designed with a wide, smooth, crescent-like separation that a polynomial function can approximate reasonably well. Its structure is globally non-linear in a way that fits a polynomial curve. In contrast, the Banknote dataset's clusters, while not perfectly linearly separable, have a more complex and irregular boundary. The data doesn't follow a simple, smooth polynomial curve. A standard polynomial kernel tries to fit a single, rigid polynomial equation across the entire feature space, which is not flexible enough to capture the nuanced, localized separation between the "Forged" and "Genuine" clusters.

PART 3:



Analysis Questions

1. Compare the two plots. Which model, the "Soft Margin" ($C=0.1$) or the "Hard Margin" ($C=100$), produces a wider margin?

After comparing the two plots, the **Soft Margin** produces a visibly wider margin. The decision boundary in the soft margin plot is positioned more centrally between the two main groups of data points, creating a larger separation between them. The Hard Margin model's boundary is pushed closer to the red data points to correctly classify all the blue ones, resulting in a narrower margin.

2. Look closely at the "Soft Margin" ($C=0.1$) plot. You'll notice some points are either inside the margin or on the wrong side of the decision boundary. Why does the SVM allow these "mistakes"? What is the primary goal of this model?

The soft margin SVM allows some data points to be misclassified or fall within the margin because its primary goal is **generalization**, not perfection on the training set. The C parameter controls the trade-off between maximizing the margin width and minimizing the classification error. A small C value (like 0.1) tells the model to prioritize a wider margin, even if it means tolerating a few misclassifications. It essentially treats these "mistakes" as potential noise or outliers, focusing instead on finding a simpler, more robust boundary that captures the overall trend of the data. The goal is to create a model that will **perform better on new, unseen data**.

3. Which of these two models do you think is more likely to be overfitting to the training data? Explain your reasoning.

The **Hard Margin** is far more likely to be overfitting to the training data. Overfitting occurs when a model learns the training data too well, including its noise and specific quirks, which hurts its ability to generalize. A large C value forces the SVM to create a hard margin by trying to classify every single data point correctly. This results in a narrow margin and a complex decision boundary that is **highly sensitive** to the exact position of every training point, including outliers. This hypersensitivity to the training data is the definition of overfitting.

4. Imagine you receive a new, unseen data point. Which model do you trust more to classify it correctly? Why? In a real-world scenario where data is often noisy, which value of C (low or high) would you generally prefer to start with?

For a new, unseen data point, I would trust the **Soft Margin** model more to classify it correctly. The reason is its **superior generalization**. Because the soft margin model is less influenced by noise and outliers in the training data, its decision boundary is more stable and more likely to represent the true underlying relationship between the classes. The hard margin model, having overfit the training data, might make poor predictions if a new point doesn't conform to the noisy patterns it learned.

In a real-world scenario where data is almost always noisy, it is generally preferable to start with a **low value** of C . A low C creates a soft margin, which provides a more robust and generalized baseline model that is less prone to overfitting from the outset. We can then tune the C parameter as needed, but starting with a bias towards generalization is a safer and more effective strategy.